

Exponents and Scientific Notation

39 points · two class periods

Name _____

Date _____

Period _____

Show your work. Circle your final answer.

Day 1

Sections 1 and 2 — numerical bases · 19 points

1. Laws of Exponents

9 points

1. Write each expression as a single power.

a. $4^3 \cdot 4^5$ 1 pt

b. $(2^3)^4$ 1 pt

c. $\frac{7^9}{7^4}$ 1 pt

d. 9^0 (give its value) 1 pt

2. Find the value of each expression.

a. $\left(\frac{2}{3}\right)^3$

1 pt

b. $6^0 \cdot 5^2$

1 pt

c. $\frac{(10^2)^2}{10^3}$

1 pt

3. Which expression is equivalent to $2^4 \cdot 2^3$?

1 pt

- ☐ A. 2^7
- ☐ B. 4^7
- ☐ C. 2^{12}
- ☐ D. 4^{12}

4. Find the value of n that makes the statement true: $\frac{5^n}{5^2} = 125$

1 pt

2. Integer Exponents

10 points

5. Rewrite each expression with positive exponents. Do not evaluate.

a. 6^{-3} 1 pt

b. $\frac{1}{(-2)^{-5}}$ 1 pt

c. $3 \cdot 10^{-4}$ 1 pt

6. Find the value of each expression.

a. 4^{-2} 1 pt

b. $(-3)^{-3}$ 1 pt

c. $\left(\frac{1}{2}\right)^{-3}$ 1 pt

d. $2^5 \cdot 2^{-7}$ 1 pt

7. Which is the value of 5^{-2} ?

1 pt

- ☐ A. $\frac{1}{25}$
- ☐ B. -25
- ☐ C. -10
- ☐ D. $\frac{1}{10}$

8. Select ALL of the expressions whose value is $\frac{1}{64}$.

1 pt

- ☐ A. 2^{-6}
- ☐ B. 4^{-3}
- ☐ C. -8^2
- ☐ D. $\left(\frac{1}{4}\right)^3$
- ☐ E. 2^{-8}

9. Marco says the value of 3^{-2} is -9 . Explain why he is wrong, and give the correct value. 1 pt

Day 2

Sections 3 and 4 — variable bases, and scientific notation · 20 points

3. Variable Bases

9 points

10. Write each expression as a single monomial.

a. $x^4 \cdot x^6$ 1 pt

b. $(y^3)^5$ 1 pt

c. $(3a^2)^3$ 1 pt

d. $\frac{15m^7}{5m^2}$ 1 pt

11. Rewrite each expression using only positive exponents.

a. b^{-4} 1 pt

b. $5x^{-2}$ 1 pt

c. $(2m^{-3}n)^2$ 1 pt

12. Which expression is equivalent to $(4x^3)^2$? 1 pt

- ☐ A. $16x^6$
- ☐ B. $8x^6$
- ☐ C. $16x^5$
- ☐ D. $4x^6$

13. How are x^2 and x^{-2} related? Explain, and give the value of each when $x=5$. 1 pt

4. Scientific Notation 11 points

14. Write each number in scientific notation.

a. 8,300,000 1 pt

b. 0.00047 1 pt

c. 60,500,000,000 1 pt

d. 0.0000000092 1 pt

15. Write each number in standard form.

a. 2.9×10^5

1 pt

b. 7.1×10^{-6}

1 pt

16. Which is 0.0036 written in scientific notation?

1 pt

- ☐ A. 3.6×10^{-3}
- ☐ B. 3.6×10^{-2}
- ☐ C. 36×10^{-4}
- ☐ D. 3.6×10^3

17. Answer each question.

a. How many times larger is 9×10^9 than 3×10^6 ?

1 pt

b. How many times larger is 4.8×10^{-3} than 1.6×10^{-7} ?

1 pt

18. Which is greater, 8.9×10^{-4} or 2.1×10^{-3} ? Explain how you know.

1 pt

19. A computer displays a number as $5.6E8$. Write the number in standard form.

1 pt